Lesson Plan

Teacher Name: Ms. Lubna Kakkengal and Mr. Dilan Devendra	**Class/Section:** Grade 5
Subject: TCD: Junior Designer Challenge	**Date: Week 9 (26 Oct to 30 Oct)**
Class Description: Grade 5 TCD class, Junior Designer Challenge Session 1 of 5 (1 hour 30 minutes). Students learn the Tinkercad basics, including shapes, holes, sizing in millimetres, align and group, by designing a simple keychain or name tag.	
TA/LSA Role (if applicable):	

Differentiated Learning Groups (Data)			
Desert Fox	**Camel**	**Oryx**	**Falcon**

Lesson Overview		
Lesson Topic	Tinkercad Basics: Keychain or Name Tag	
Standard(s)	Math: 5.MD.C.3 (volume and 3D figures), 5.G.A (coordinate plane); Technology: CAD basics	
Learning Objective(s) (WALT): We are learning to use the basic Tinkercad tools to design an accurate keychain or name tag.		**Success Criteria (WILF):** I can drag, resize and rotate shapes using millimetre measurements. I can use a hole shape to cut a ring hole. I can align and group shapes into one object. I can add raised text to my design.
Literacy Strategies	Tinkercad tool cards with picture and word; partners explain each tool using: This tool lets me ...	
Assessment	Saved Tinkercad design checked against the size checklist; observation; live quiz on the TCD website.	
Targeted Vocabulary	Tinkercad, workplane, millimetre, solid, hole, align, group, dimensions	
Skills	Digital design, measuring, precision, following instructions	
National Identity Integration	Students can add a simple pattern inspired by Emirati Sadu weaving or the shape of a dhow to their keychain.	
AI/Digital Learning	Tinkercad accounts through the class code; hook video; live quiz	

	on the TCD website. Presentation: Grade5_Week9_Presentation.pptx. Worksheet: Grade5_Week9_Worksheet.

Lesson Structure	
Starter (Hook) Engage and Ask (10 minutes)	Warm-up: Guess the Size (slide 4). Show the hook video and ask the hook question (slide 2). Share the learning goals (slide 3) and key words (slide 5).
I Do (20 minutes)	Tour the Tinkercad screen (slide 6). Model solids and holes (slide 7) and typing sizes in millimetres (slide 8). Model align, group and the text shape (slides 9 and 10).
We Do (15 minutes)	Students log in and follow along to make a practice box with a hole. Teacher checks that every student can type a size and group a hole.
You Do (35 minutes)	Students design their keychain or name tag (slide 12). Challenge levels on the board (slide 13). Live quiz on tablets (slide 14).
Differentiated Instruction: **Beginning / On / Above**	**Beginning: Follow the step cards to build a rectangle keychain with your initials.** **On: Design a keychain with your full name and a ring hole, all sizes correct.** **Above: Add a shape or pattern (for example a dhow or a Sadu-inspired border) and round the corners.**
Closure/ Reflection (10 minutes)	Three students share their screen and explain one tool. Exit ticket (slide 15). Check every design is saved and named.

Inclusive Planning
Students receiving additional inclusive support (IEP or ILPs) or Gifted and Talented: EAL: tool picture cards and pre-taught vocabulary. SEN: step cards, one step at a time, with teacher check-ins. G&T: add a heritage-inspired pattern and rounded corners.
Inclusion / EAL Support

☑ Visual Supports
☑ Chunked Instructions
☐ Reduced Language Load
☑ Model / Worked Example
☑ Pre-teach Vocabulary
☐ TA/LSA Support
☐ Extra Processing Time
☐ Text Read by TA/LSA
☐ Flexible Grouping
☐ Assistive Technology
☐ Movement Break
☐ Scaffold / Sentence Starters
☐ Other:

G&T / Extension

☑ Extension / Challenge Task
☐ Higher-order Questioning
☐ Depth & Complexity Task
☐ Independent Inquiry
☐ Leadership / Peer Coaching
☐ Real-World Application
☐ Curriculum Compacting
☐ Above-grade-level Texts
☐ Student-led / Passion Project
☑ Open-Ended Task
☐ Other:

Assessment Adaptations

☑ Oral Response
☐ Reduced Questions
☐ Scaffolded Success Criteria
☐ Extra Time
☐ Alternative Outcome Format
☑ Teacher Conferencing/Live Feedback
☐ Quiet / Separate Space
☐ Other: